

Ex08. Introduction to Signals and systems

s1320112/ Motohisa YAMANO

1. Plot Transfer Functions and Response Signals

This assignment involves creating the 3d figure of the transfer function $H(z)$ and response signal $h[n]$ and $s[n]$. For the numerical computation, the discrete time index n ranges from 0 to 10 with an increment of 0.1. The frequency domain ω is sampled at intervals of 0.1 (from $-\pi$ to π). The resulting waveforms are organized into a 2×2 subplot grid for comparative analysis.

```
r = 0:0.01:1;
omega1 = -pi:0.01:pi;
omega2 = -pi:0.1:pi;
n = 0:0.1:10;

H = @(z) (z.^2) ./ (z.^2 - (3/4)*z + (1/8));
h = @(n) (2 * (1/2).^n - (1/4).^n) .* (n >= 0);
s = @(n) ((8/3) + (1/3)*((1/4).^n) - 2 * ((1/2).^n)) .* (n >= 0);

figure('Position', [100, 100, 800, 800]);

subplot(2, 2, 1);
[R, W] = meshgrid(r, omega1);
Z = R .* exp(1j * W);
p1 = plot3(real(Z), imag(Z), 20*log10(abs(H(Z))), 'b');
title('1. plot3 Transfer Function H(z)');
xlabel('Real(z)');
ylabel('Imag(z)');
zlabel('20log_{10}|H(z)|');
grid on;

subplot(2, 2, 2);
[R, W] = meshgrid(r, omega2);
Z = 1 .* exp(1j * W);
p2 = stem(omega2, abs(H(Z)), 's-');
title('2. stem Transfer Function H(z)');
xlabel('\omega');
ylabel('|H(e^{j\omega})|');
grid on;

subplot(2, 2, 3);
p3 = stem(n, h(n), 's-');
title('3. stem Impulse Response h[n]');
xlabel('n');
ylabel('h[n]');
grid on;

subplot(2, 2, 4);
```

```

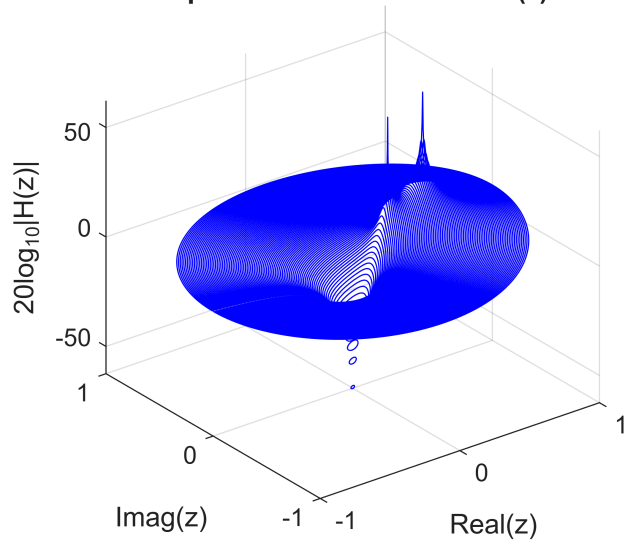
p4 = stem(n, s(n), 's-');
title('4. stem Step Response s[n]');
xlabel('n');
ylabel('s[n]');
grid on;

sgtitle("Transfer Functions and Response Signals");

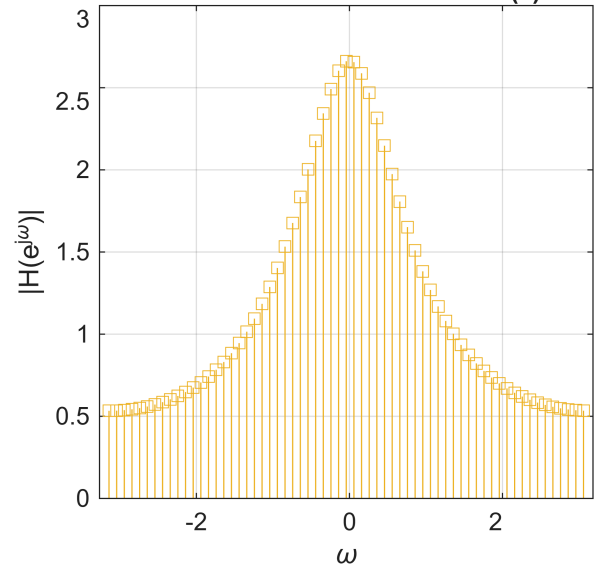
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Transfer Functions and Response Signals

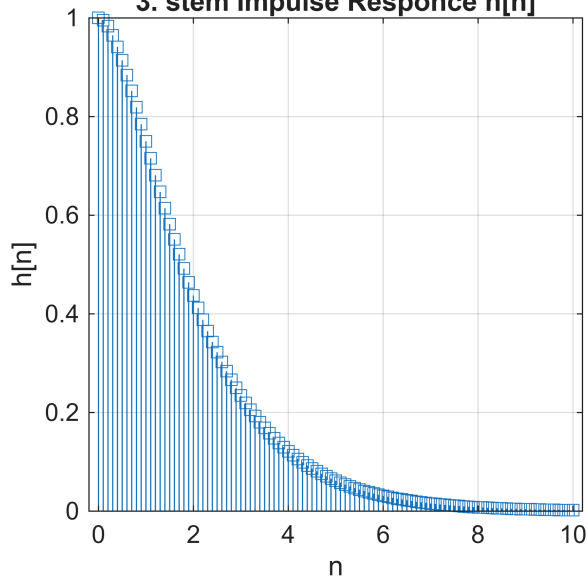
1. plot3 Transfer Function $H(z)$



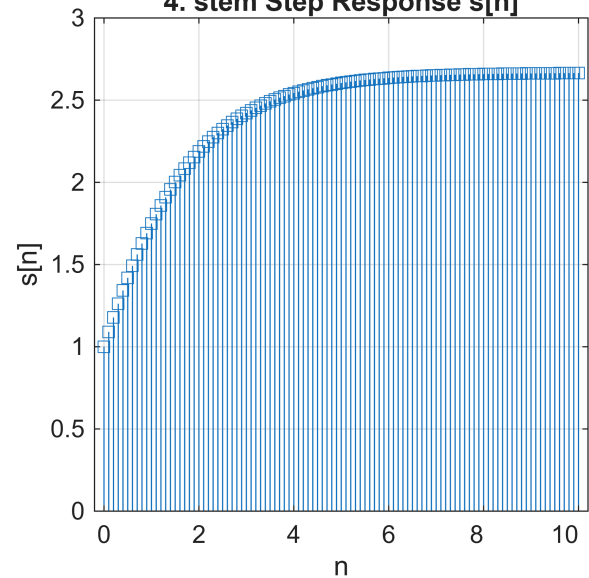
2. stem Transfer Function $H(z)$



3. stem Impulse Response $h[n]$



4. stem Step Response $s[n]$



2. Math Problem

Below is the snapshot of my answer in fill-blank math problem.

$$Y(z) - \frac{3}{4}z^{-1}Y(z) + \frac{1}{8}z^{-2}Y(z) = X(z)$$

$$\frac{1}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}, \quad \frac{z^2}{z^2 - \frac{3}{4}z + \frac{1}{8}}, \quad \frac{z^2}{(z - \frac{1}{4})(z - \frac{1}{2})}$$

$$\frac{z}{(z - \frac{1}{4})(z - \frac{1}{2})}, \quad C_1 = -1, \quad C_2 = 2, \quad H(z) = \frac{-1}{1 - \frac{1}{4}z^{-1}} + \frac{2}{1 - \frac{1}{2}z^{-1}}$$

$$h[n] = -\left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{2}\right)^n u[n], \quad Y(z) = \frac{z^3}{(z-1)(z-\frac{1}{4})(z-\frac{1}{2})}$$

$$\frac{Y(z)}{z} = \frac{\frac{8}{3}}{z-1} + \frac{\frac{1}{3}}{z-\frac{1}{4}} - \frac{2}{z-\frac{1}{2}}, \quad C_1 = \frac{8}{3}, \quad C_2 = \frac{1}{3}, \quad C_3 = -2$$

$$s[n] = \left(\frac{8}{3} + \frac{1}{3}\left(\frac{1}{4}\right)^n - 2\left(\frac{1}{2}\right)^n\right) u[n]$$